



中山大學

SUN YAT-SEN UNIVERSITY

第九章 移位寄存器

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计算机学院

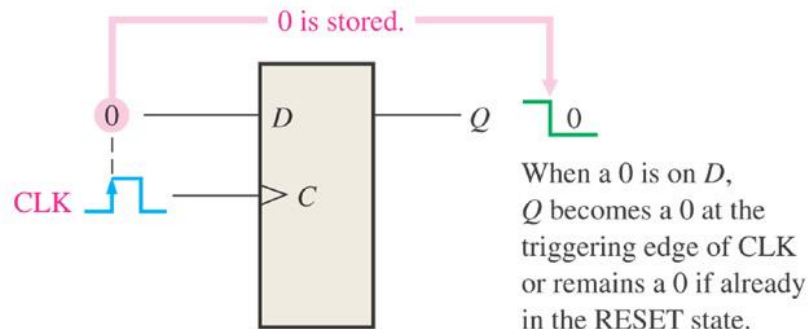
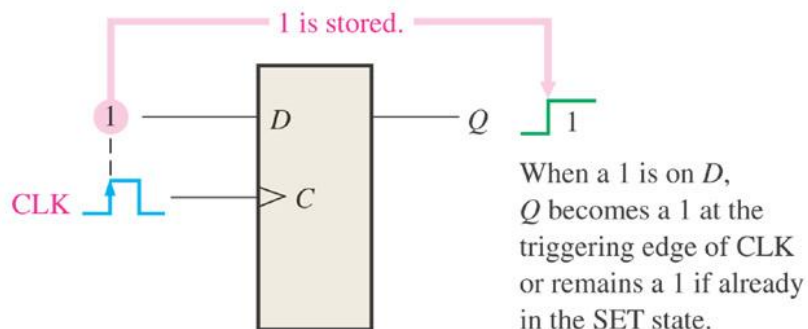
2026年5月13日



概述

• 寄存器：

- 两个基本功能：数据存储、数据移动。
- 一个或多个触发器用于存储和移位数据





概述

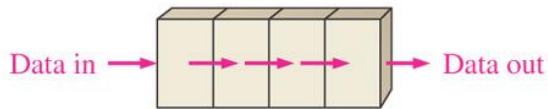
- 移位功能：

- 指寄存器里存储的代码，能在移位脉冲的作用下依次左移或右移。
- 移位寄存器不但可以用来寄存代码，还可以用来实现数据的串行-并行转换、数值的运算及数据处理等。

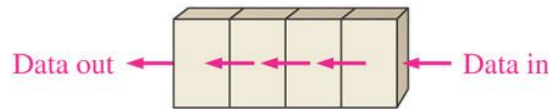


移位寄存器

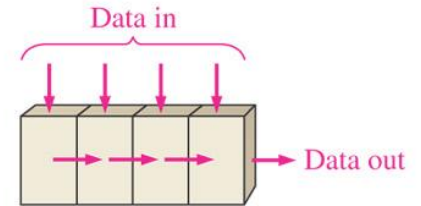
- 基本移位寄存器的功能



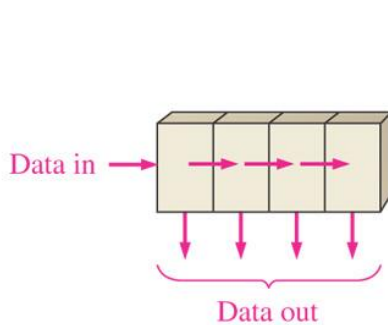
(a) Serial in/shift right/serial out



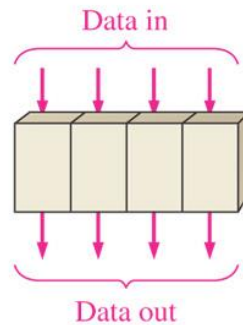
(b) Serial in/shift left/serial out



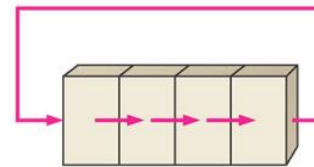
(c) Parallel in/serial out



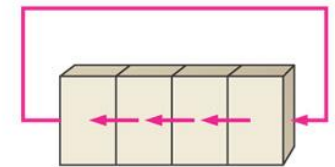
(d) Serial in/parallel out



(e) Parallel in/parallel out



(f) Rotate right

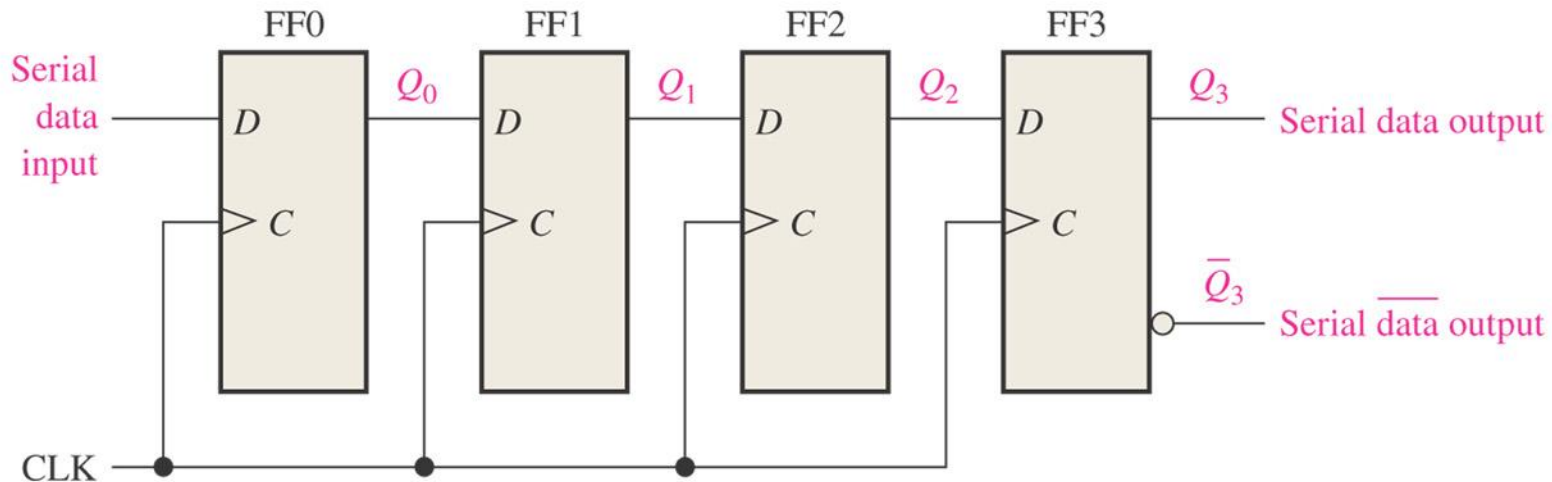


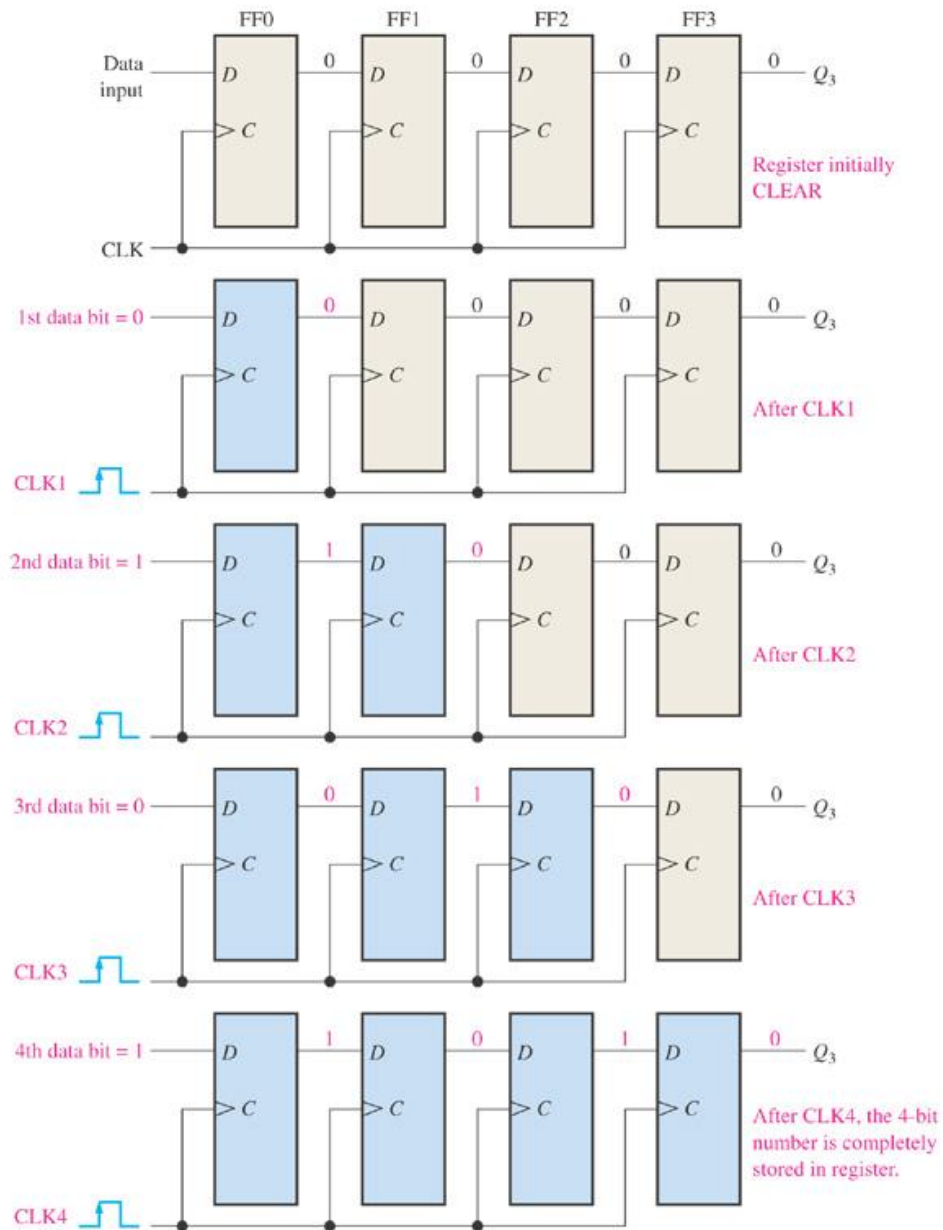
(g) Rotate left



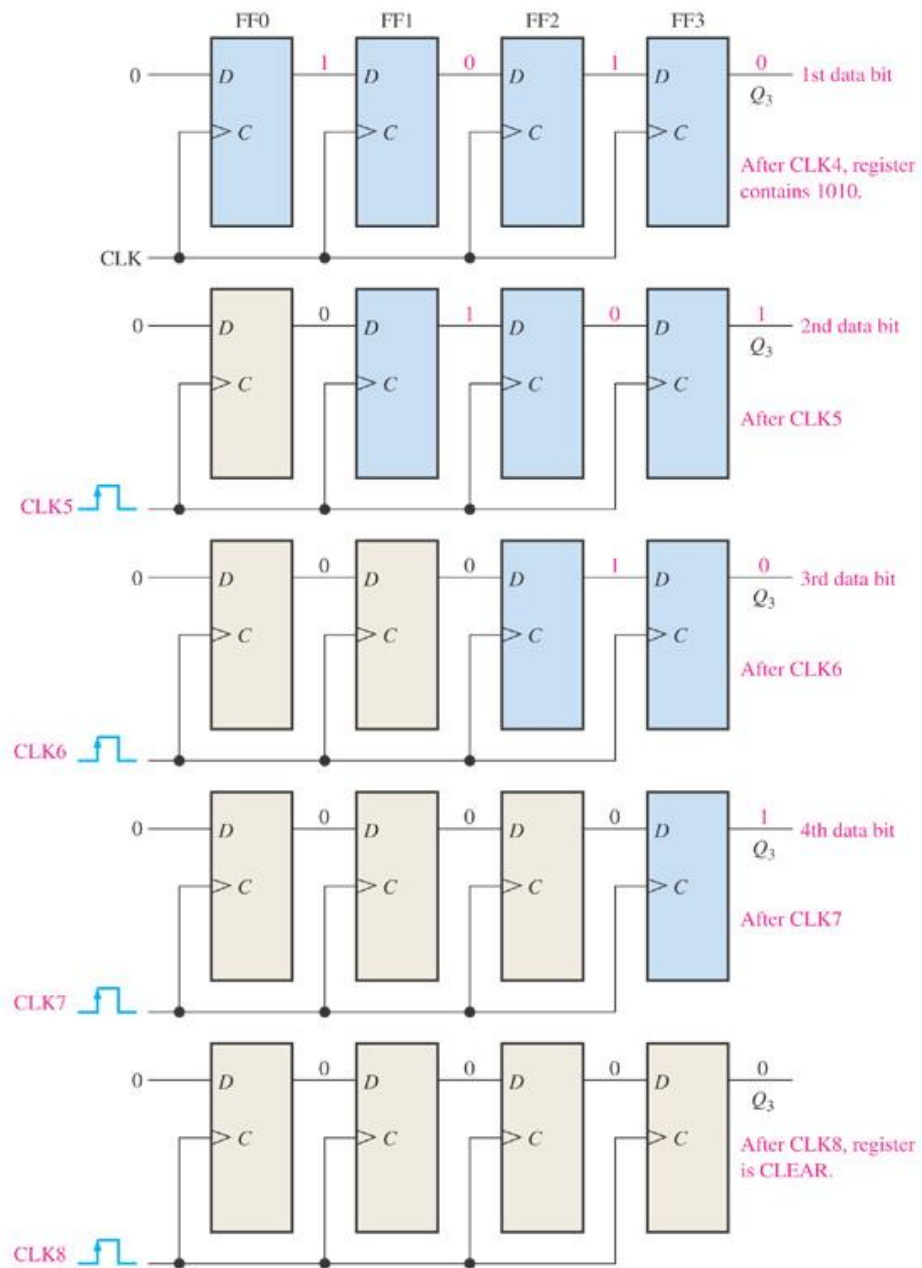
移位寄存器

- 串行输入/串行输出移位寄存器：
 - 接收串行数据
 - 一根数据线上一次输入一位
 - 在其输出上也以串行形式生成存储的信息



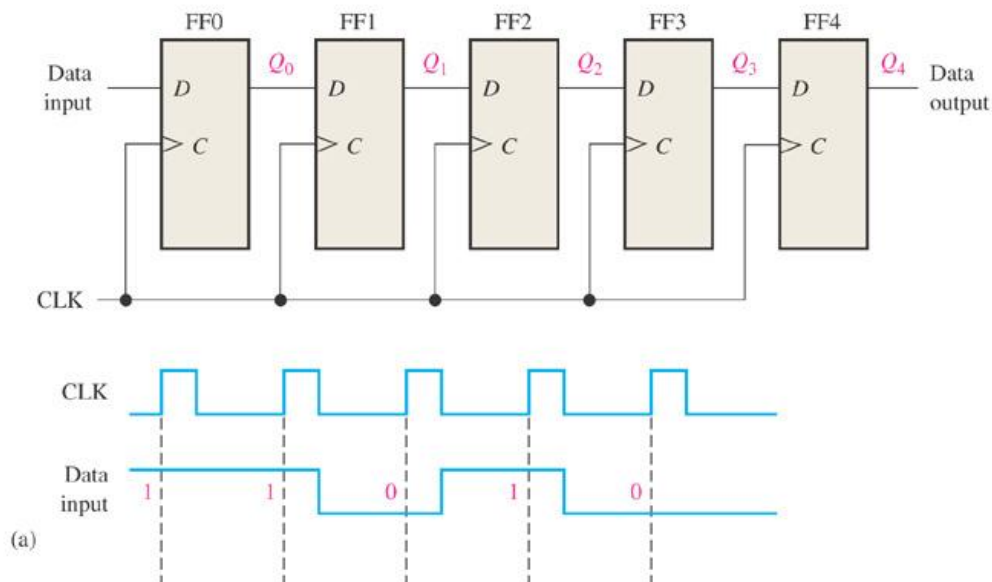


4个位串行进入寄存器



4个位串行输出寄存器，然后由0替代

练习：对于指定的数据和时钟波形，给出5位寄存器的状态
(假定该寄存器初始时刻被清零)



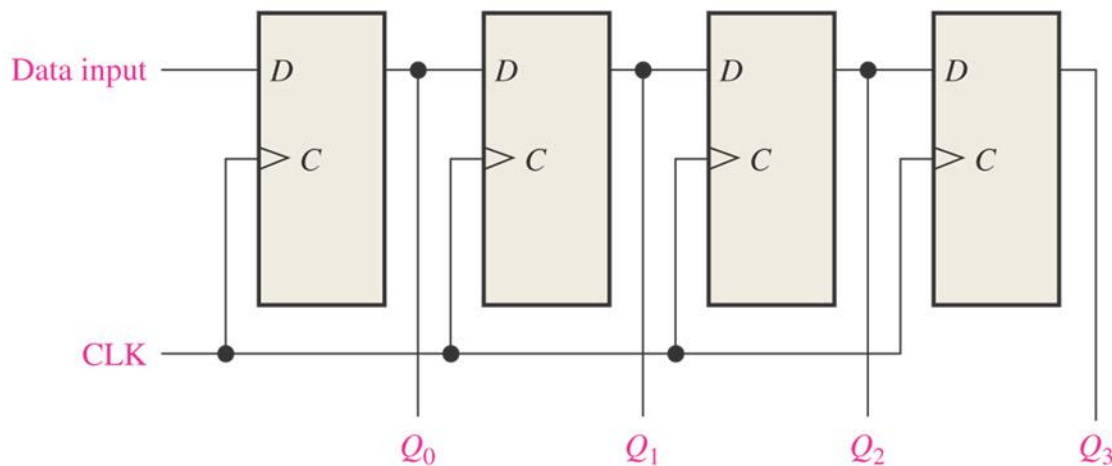
(b)



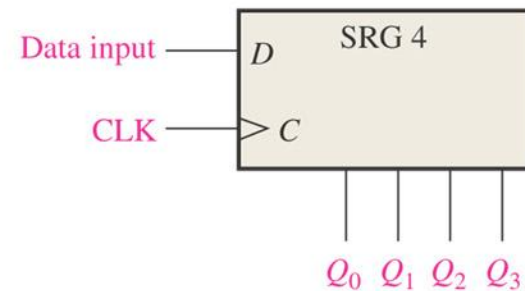
移位寄存器

- 串行输入/并行输出移位寄存器：

- 数据位串行进入寄存器
- 数据位以并行方式从寄存器中取出



(a)

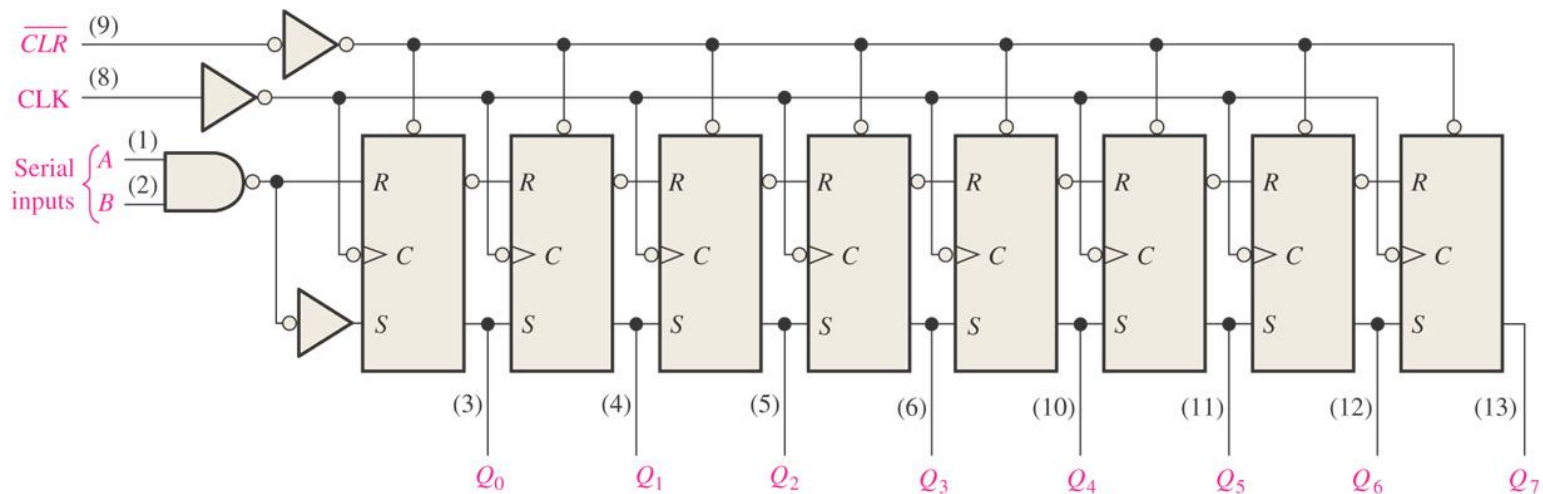


(b)

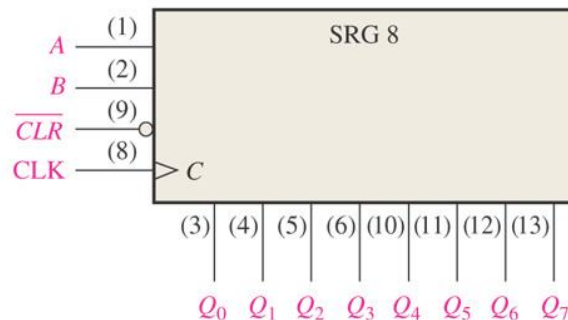


移位寄存器

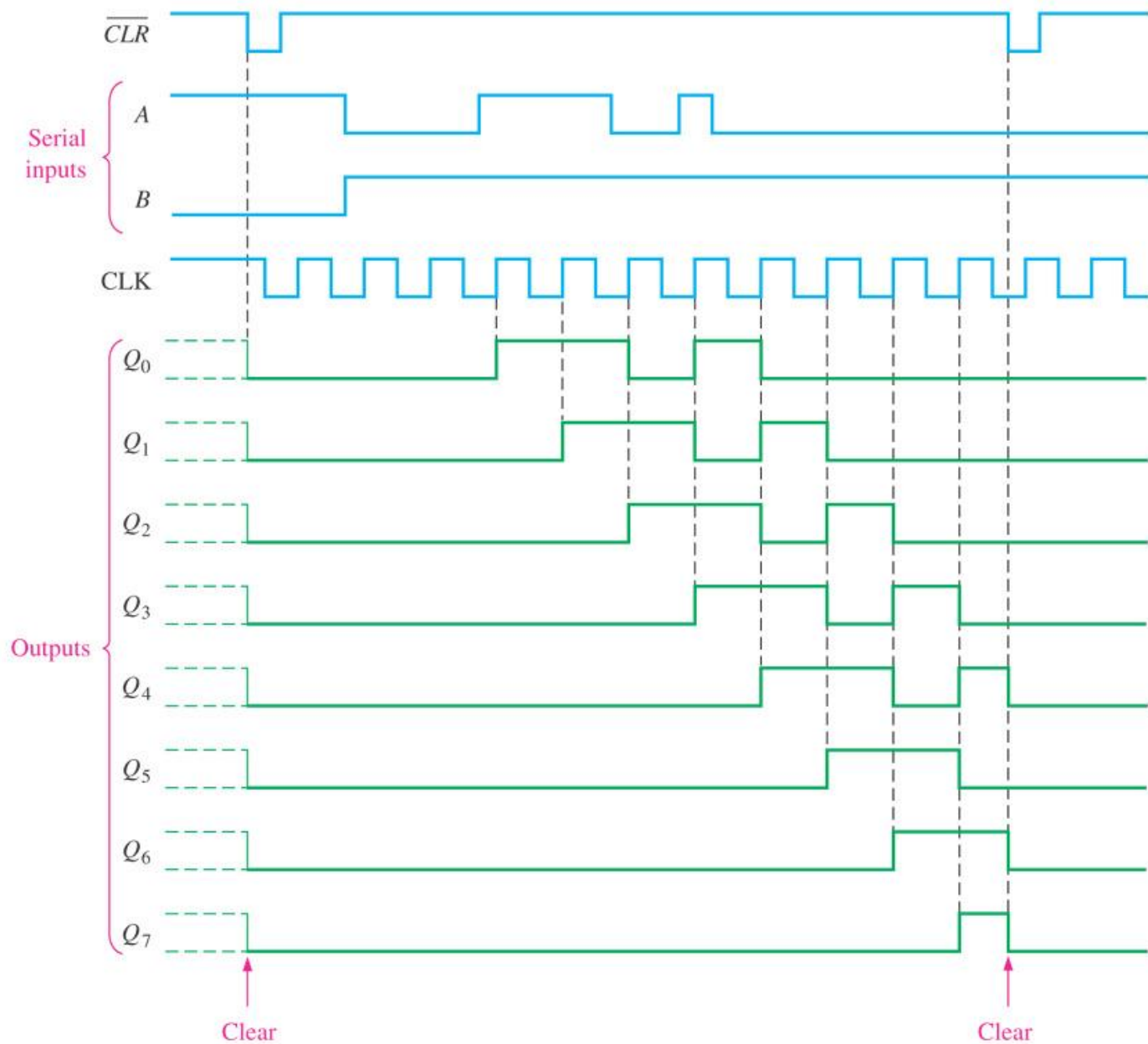
- 74HC164 8位串行输入/并行输出移位寄存器：



(a) Logic diagram



(b) Logic symbol

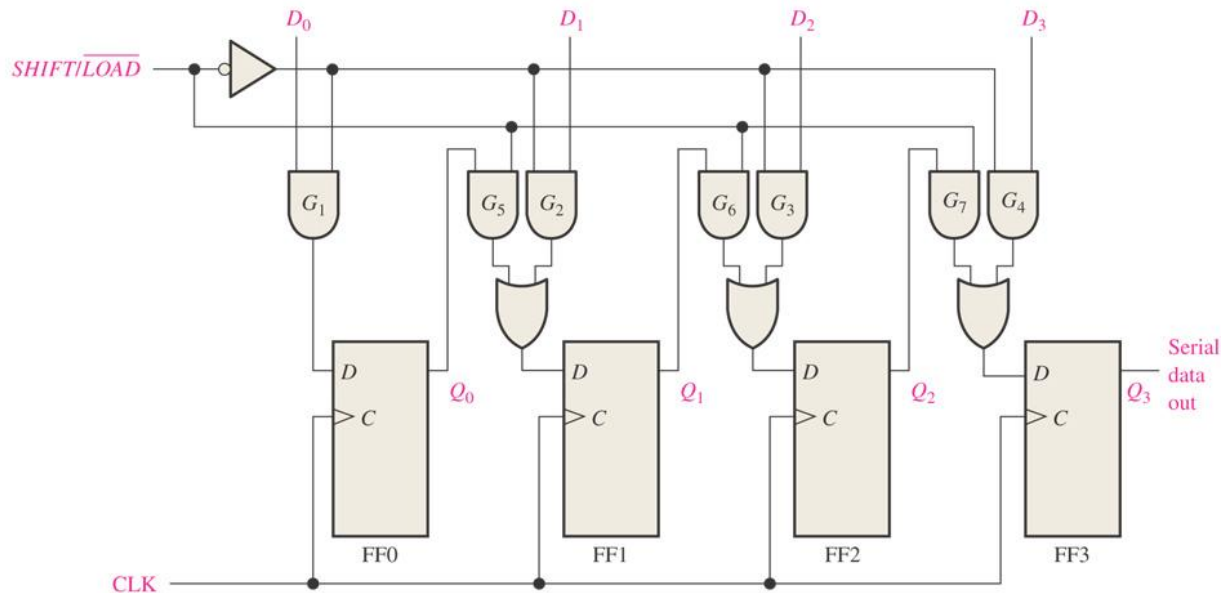


74HC164移位寄存器的样例时序图



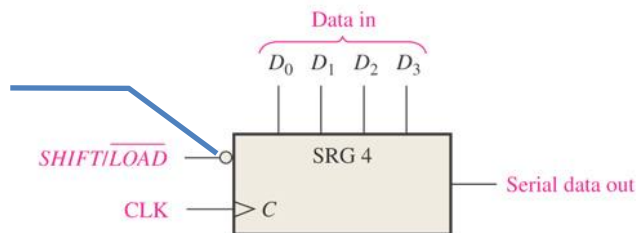
移位寄存器

- 并行输入/串行输出移位寄存器：



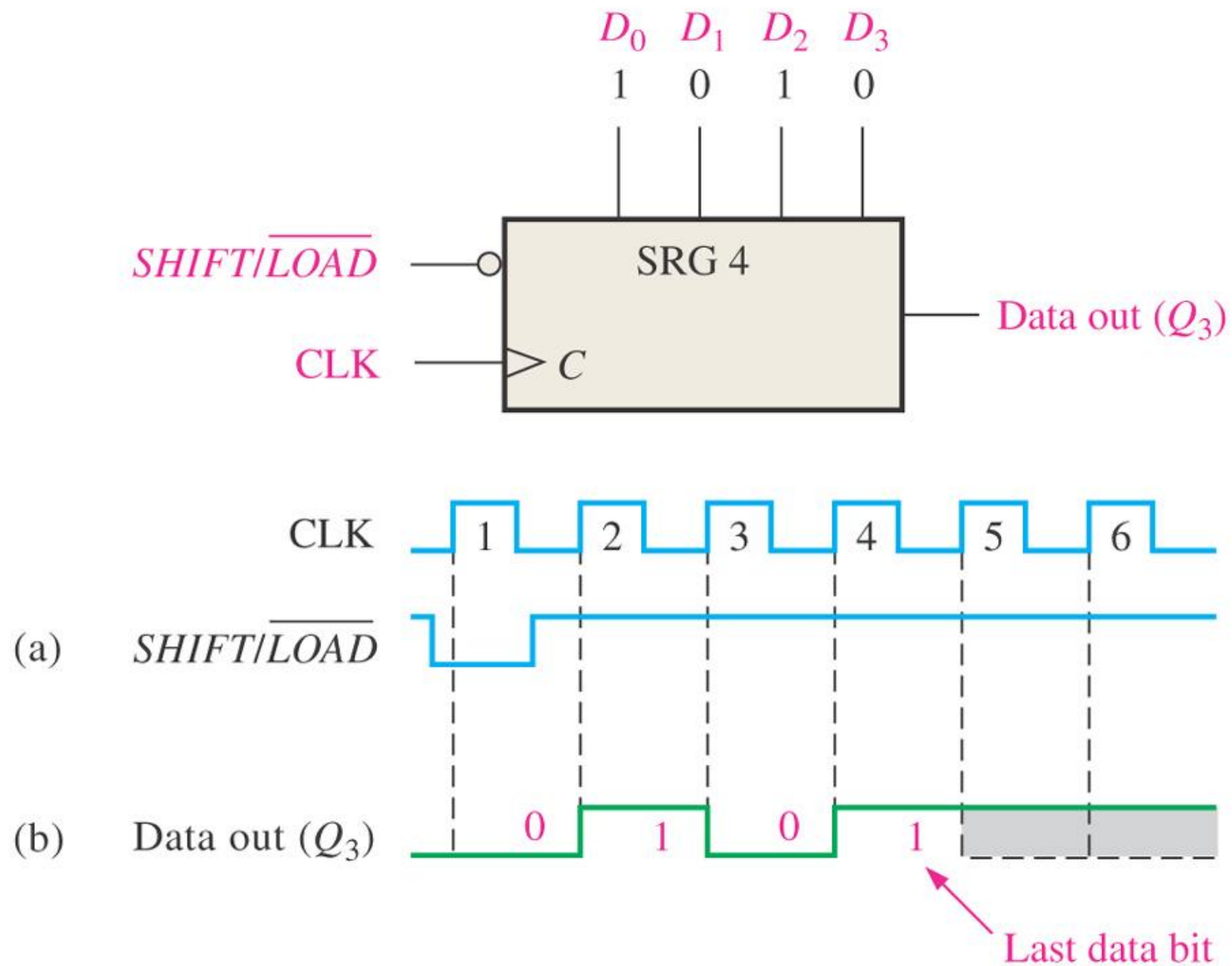
(a) Logic diagram

低电平： G_1 - G_4 开启，并行存储 D_0 - D_3
高电平： G_5 - G_7 开启，数据向右移位



(b) Logic symbol

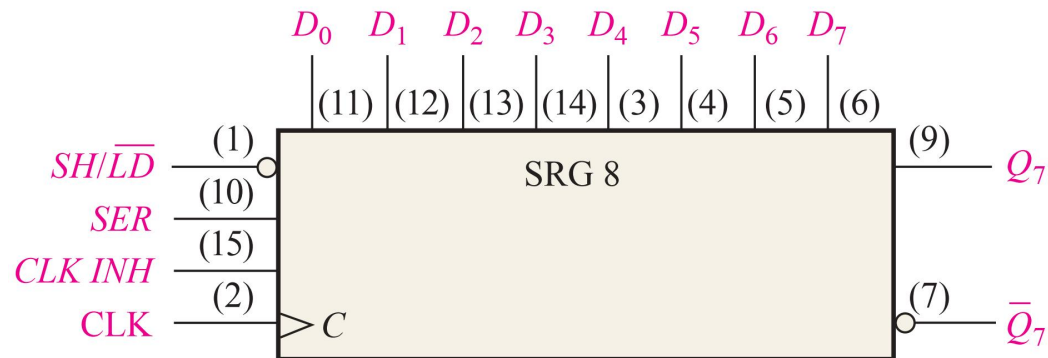
例题：给出寄存器的输出波形

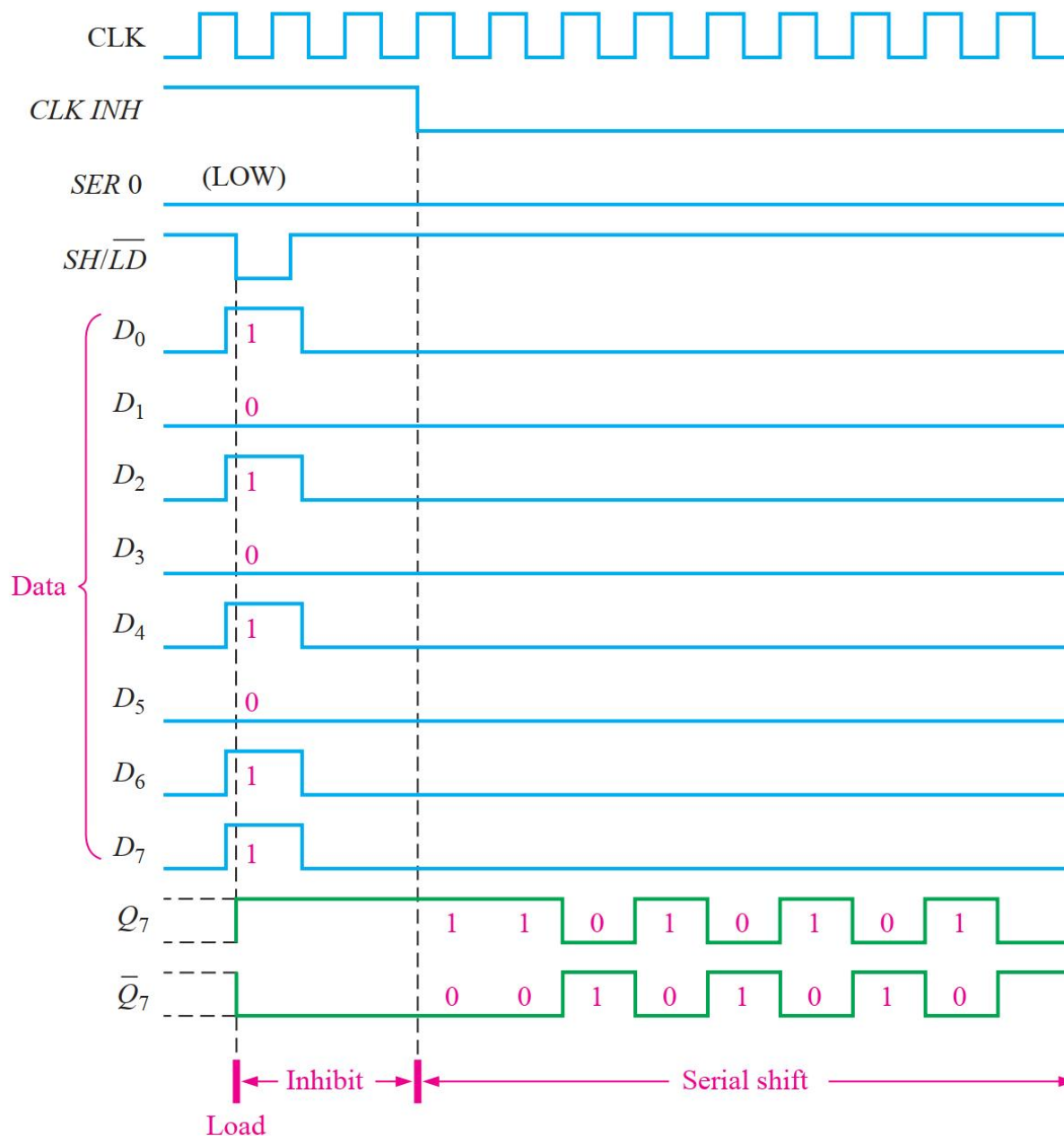




移位寄存器

- 74HC165 8位并行置数移位寄存器：



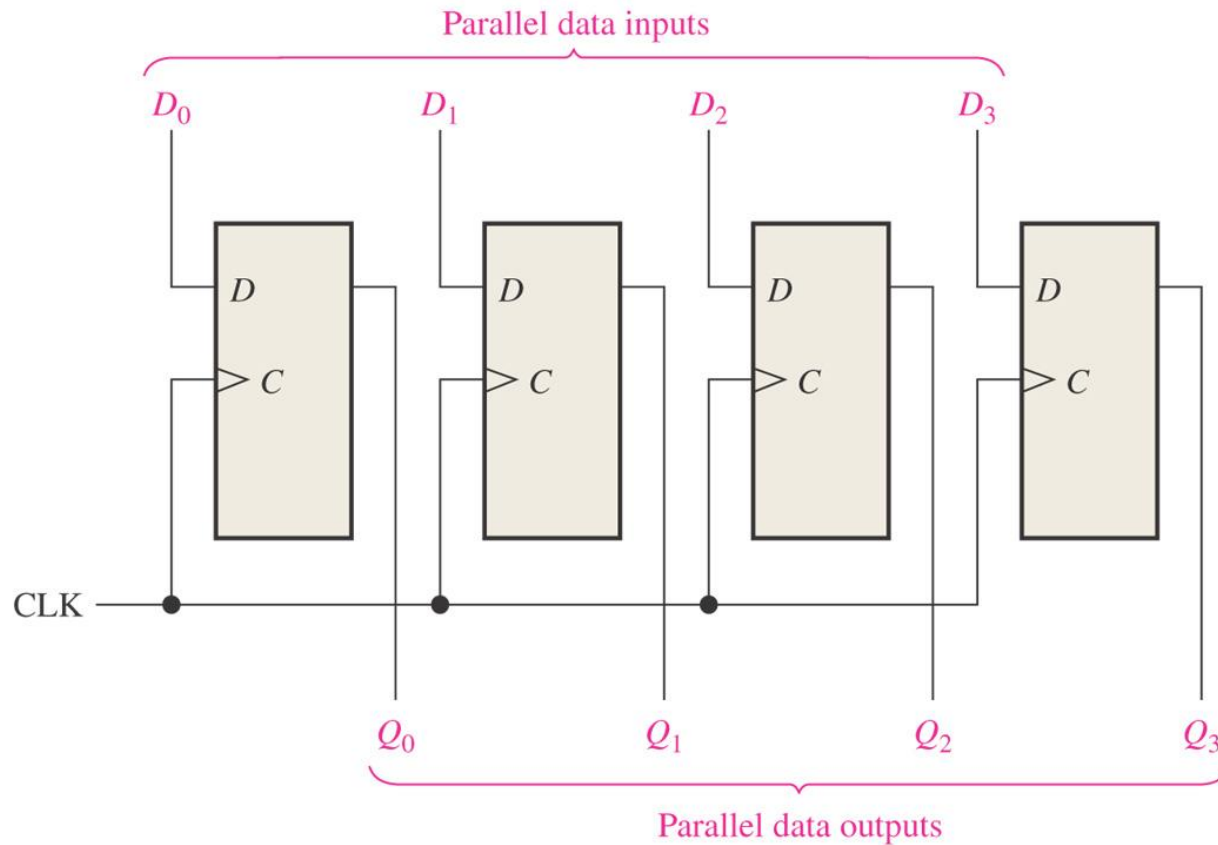


74HC165移位寄存器的样例时序图



移位寄存器

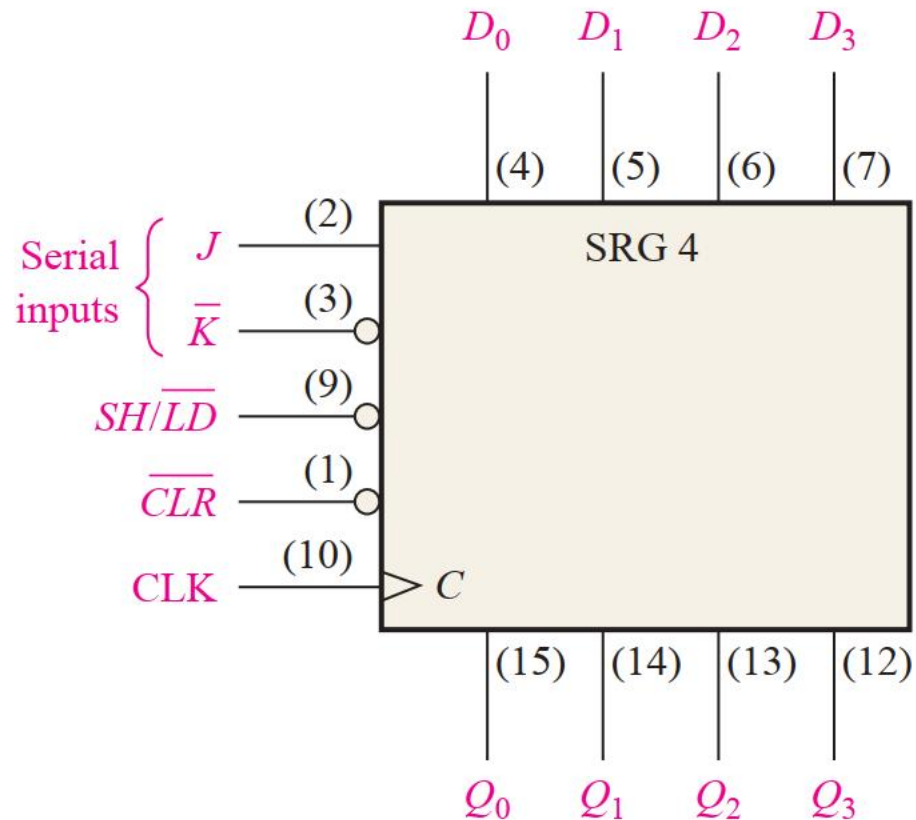
- 并行输入/并行输出移位寄存器：

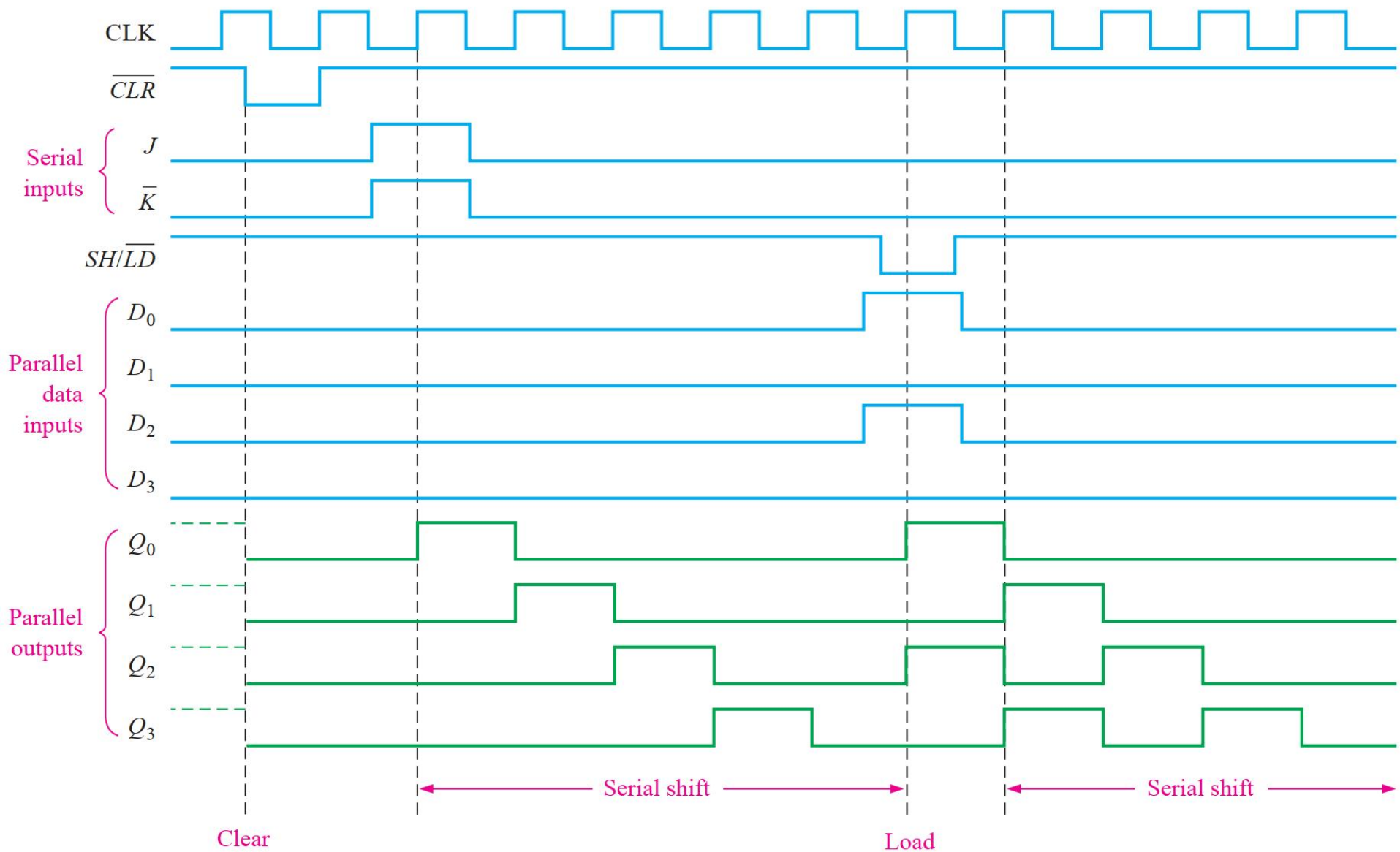




移位寄存器

- 74HC195 4位并行存取移位寄存器：



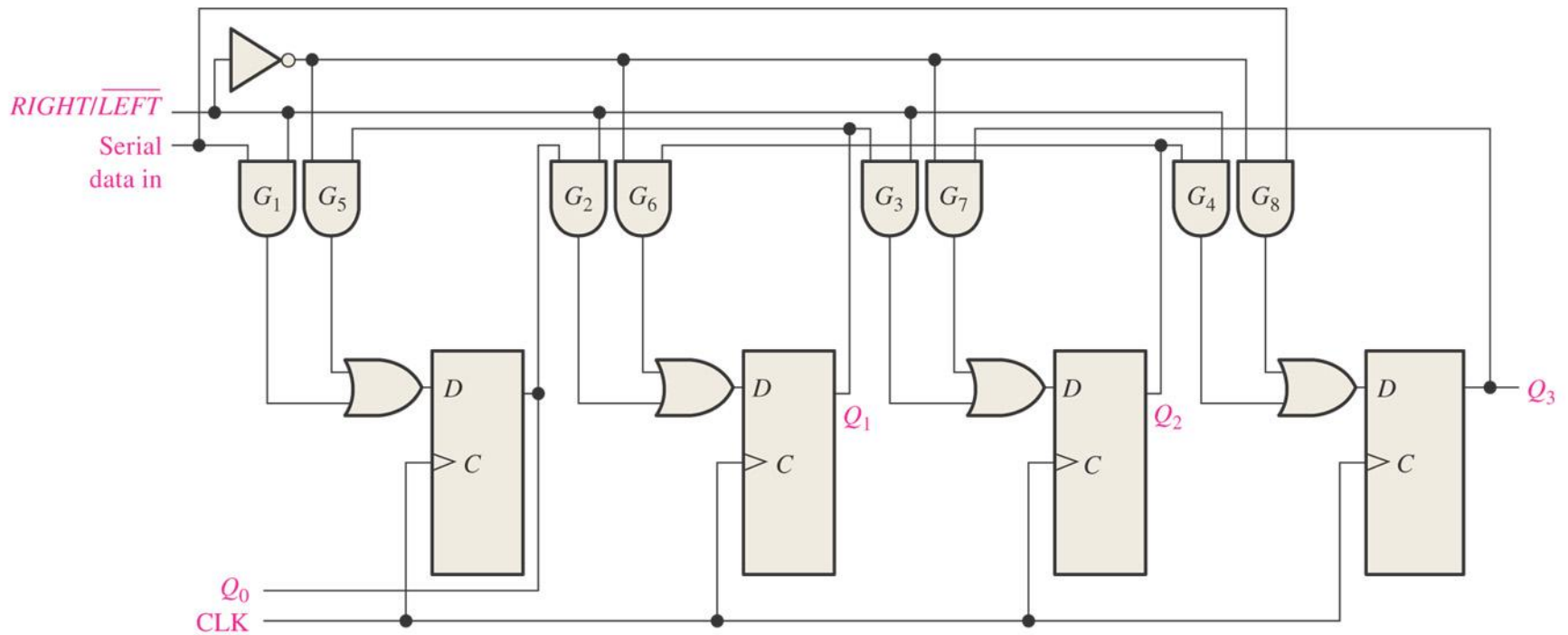


74HC195移位寄存器的样例时序图



移位寄存器

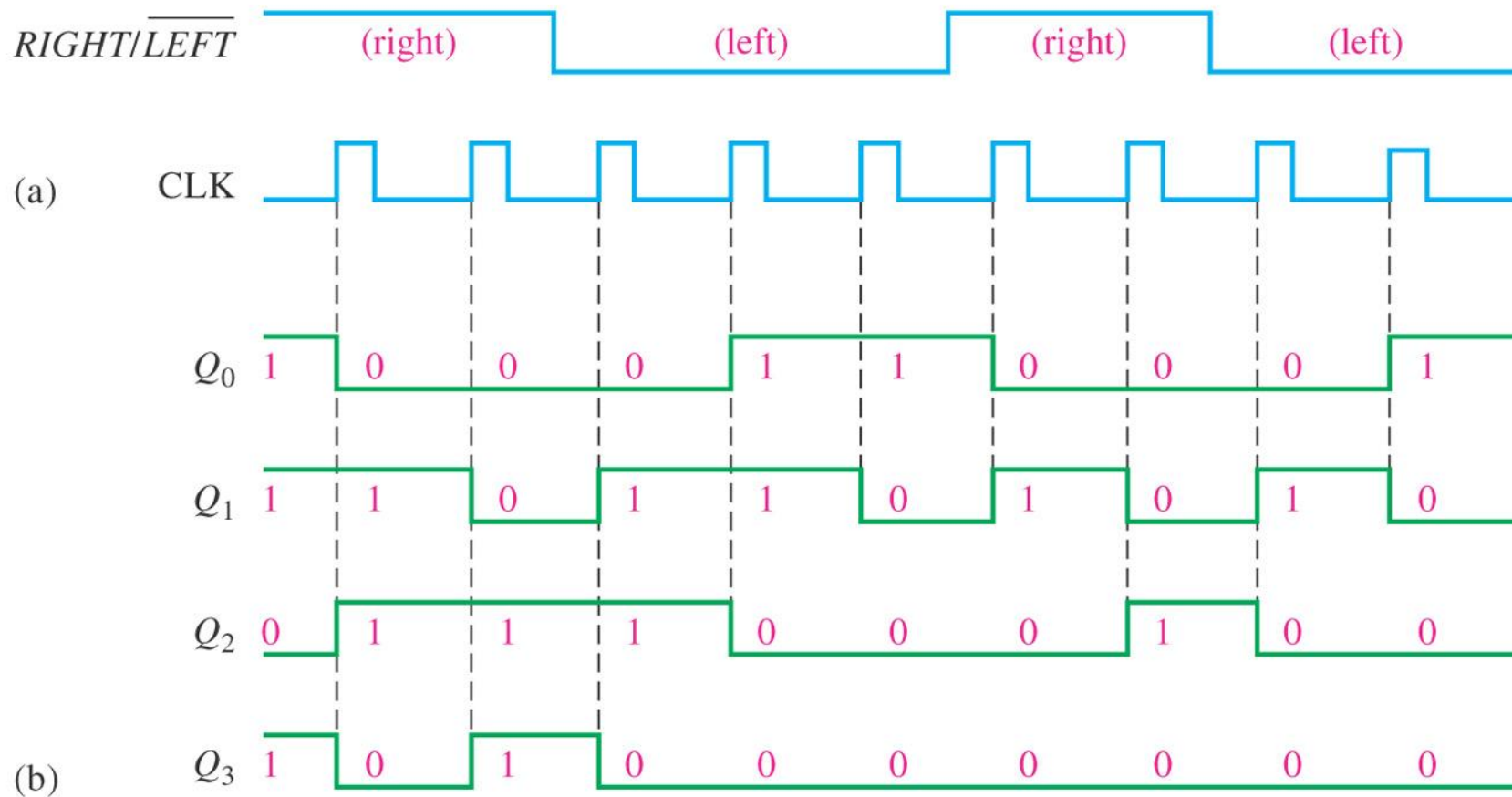
- 双向移位寄存器：





移位寄存器

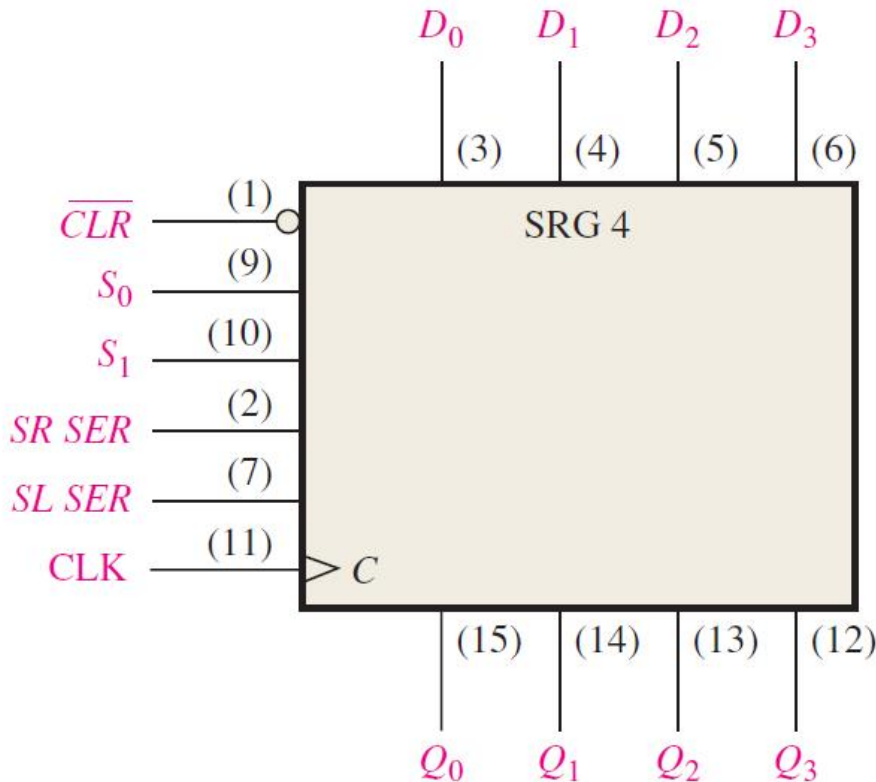
- 双向移位寄存器:



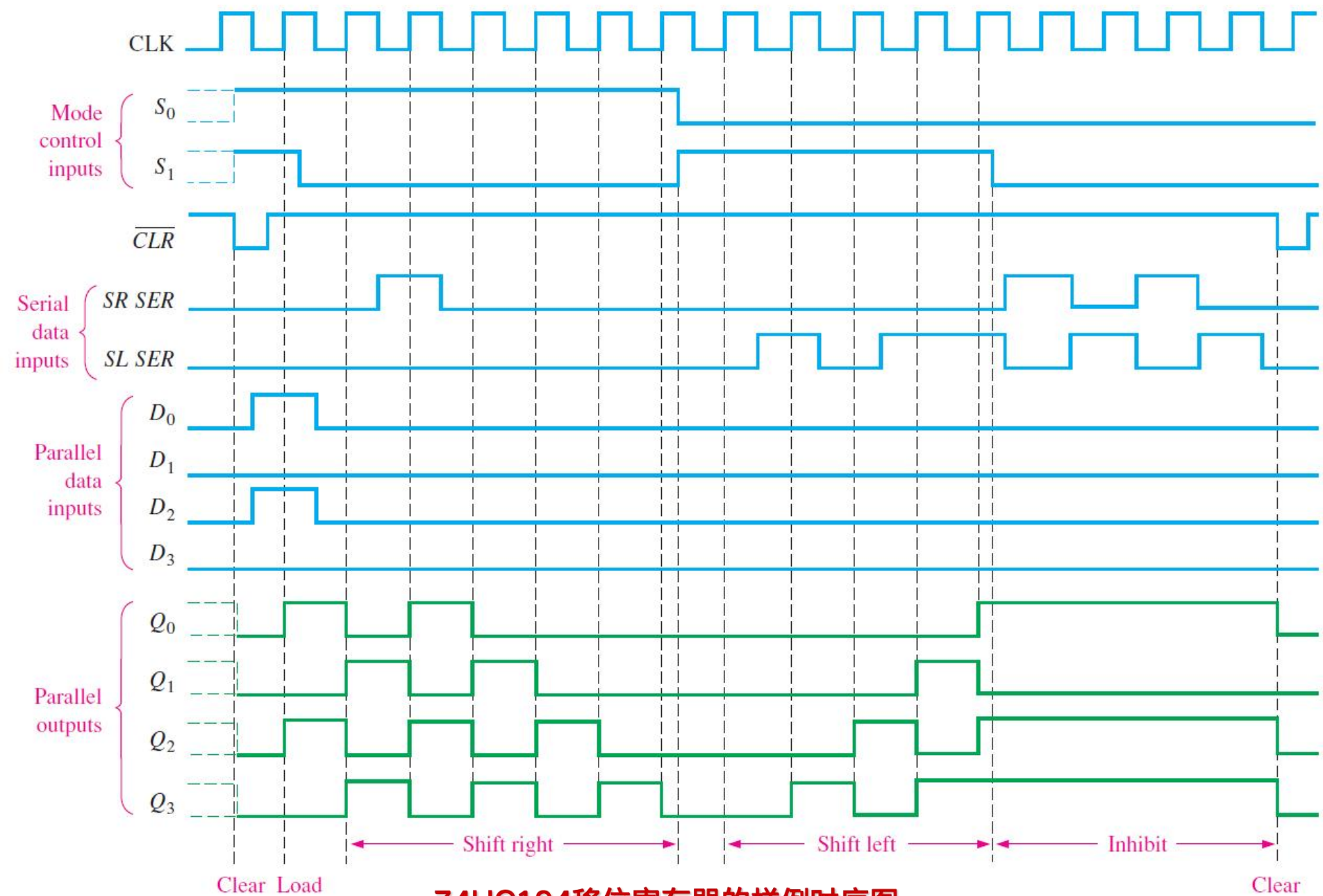


移位寄存器

- 74HC194 4位通用双向移位寄存器：



- $S_0=S_1=1$: Data Preset
- $S_0=1, S_1=0$: Shift Right, Input from SR SER
- $S_0=0, S_1=1$: Shift Left, Input from SL SER
- $S_0=S_1=0$: Inhibit

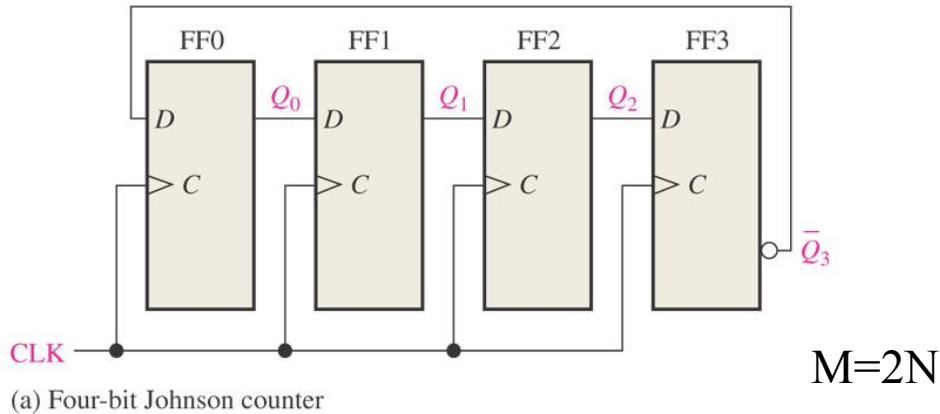


74HC194移位寄存器的样例时序图



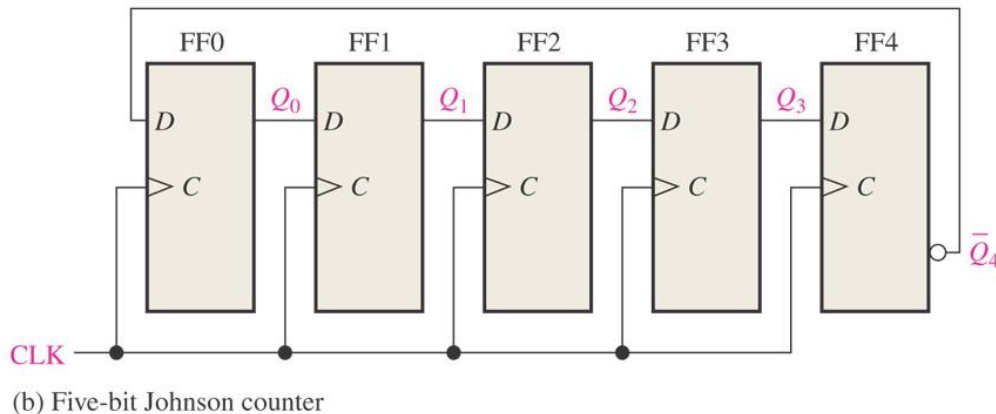
移位寄存器

• 移位寄存器计数器：约翰逊计数器



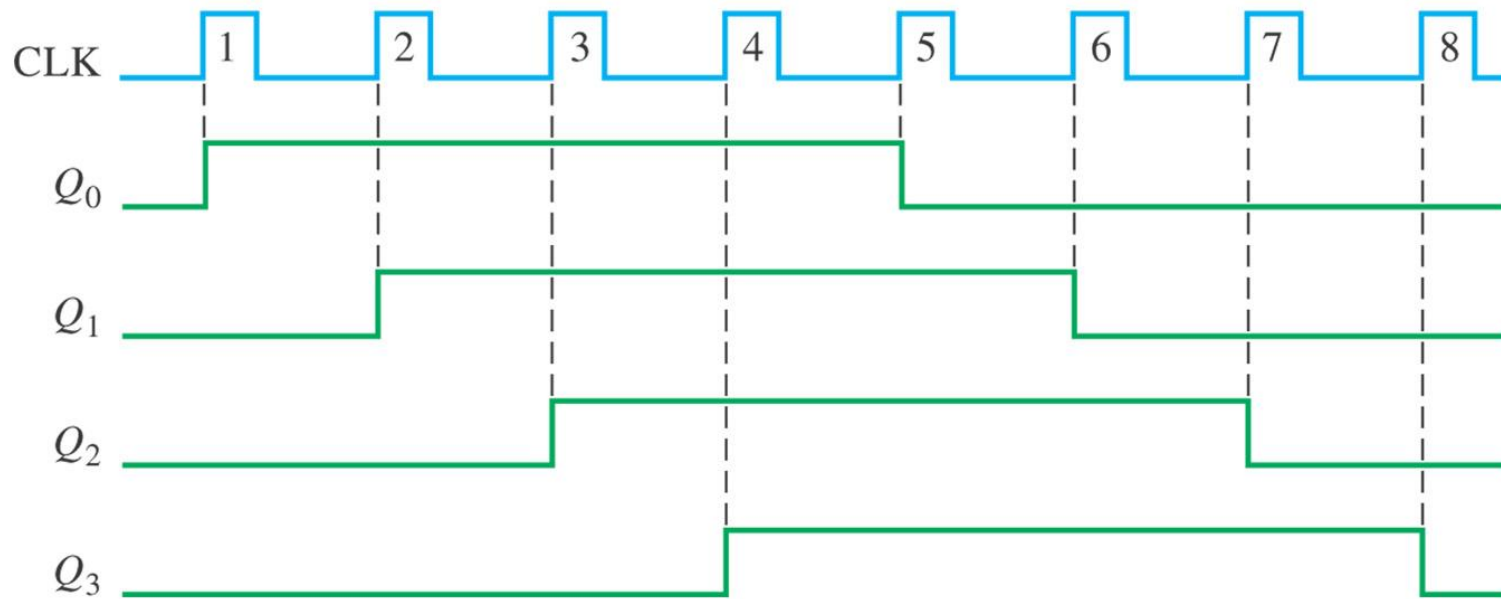
Four-bit Johnson sequence.

Clock Pulse	Q_0	Q_1	Q_2	Q_3
0	0	0	0	0
1	1	0	0	0
2	1	1	0	0
3	1	1	1	0
4	1	1	1	1
5	0	1	1	1
6	0	0	1	1
7	0	0	0	1

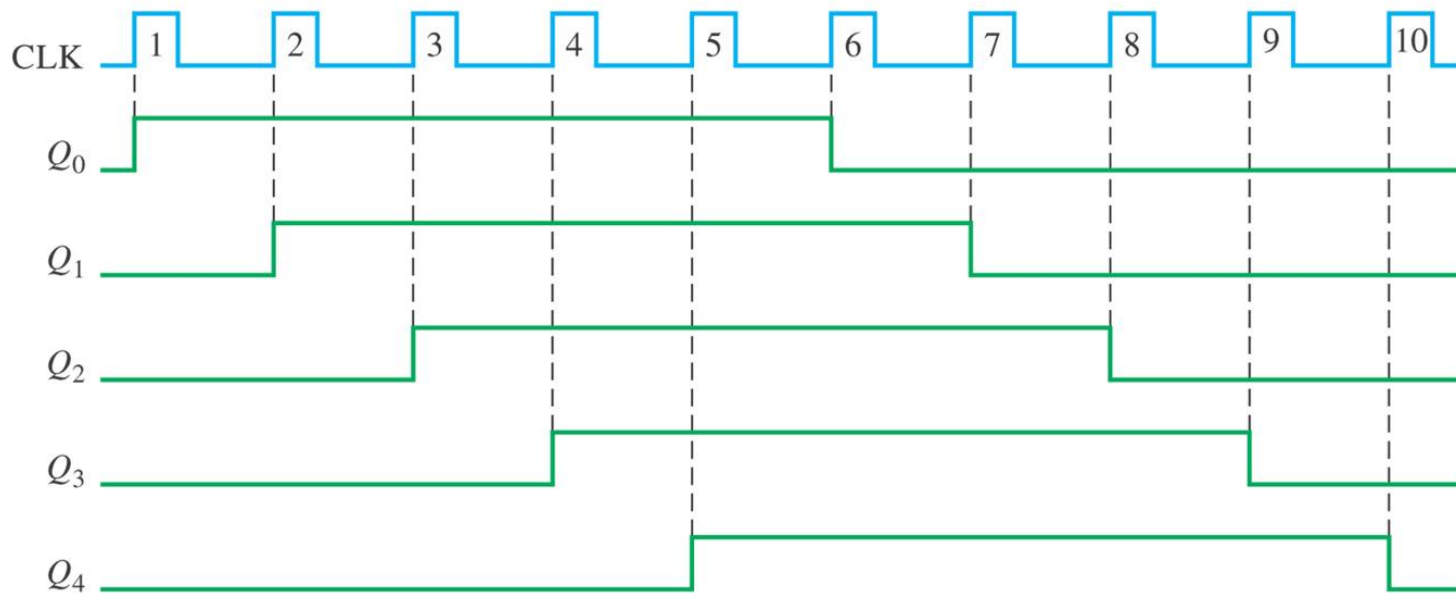


Five-bit Johnson sequence.

Clock Pulse	Q_0	Q_1	Q_2	Q_3	Q_4
0	0	0	0	0	0
1	1	0	0	0	0
2	1	1	0	0	0
3	1	1	1	0	0
4	1	1	1	1	0
5	1	1	1	1	1
6	0	1	1	1	1
7	0	0	1	1	1
8	0	0	0	1	1
9	0	0	0	0	1



4位约翰逊计数器的时序图

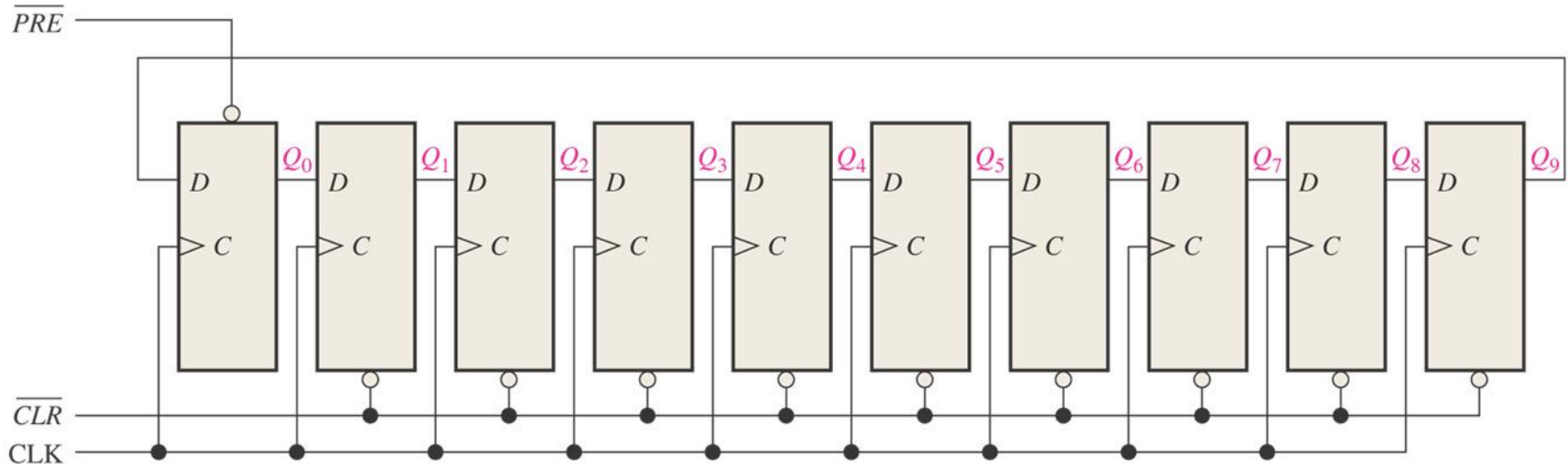


5位约翰逊计数器的时序图



移位寄存器

- 移位寄存器计数器：环形计数器



10位环形计数器



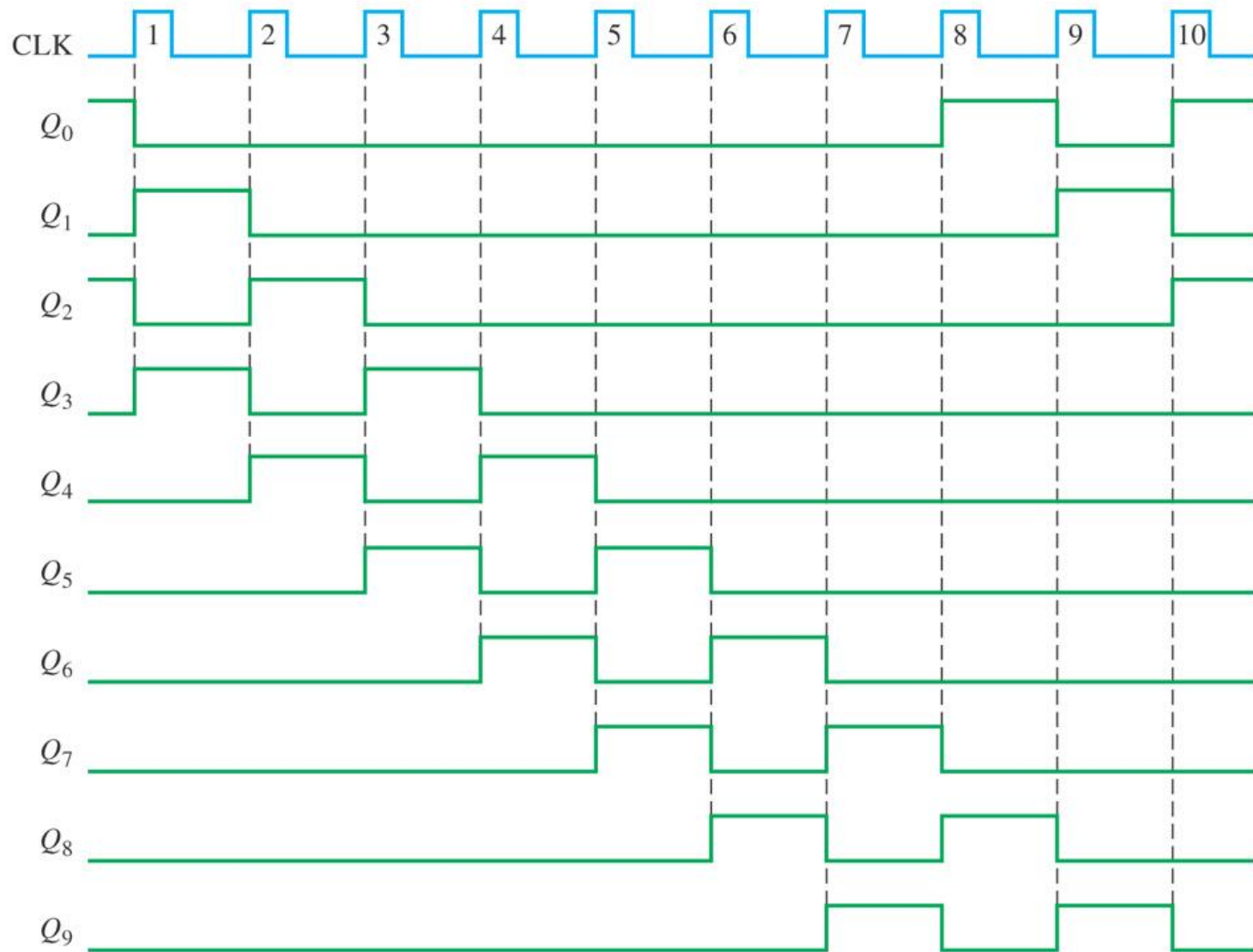
移位寄存器

- 移位寄存器计数器：环形计数器

Ten-bit ring counter sequence.

Clock Pulse	Q_0	Q_1	Q_2	Q_3	Q_4	Q_5	Q_6	Q_7	Q_8	Q_9
0	1	0	0	0	0	0	0	0	0	0
1	0	1	0	0	0	0	0	0	0	0
2	0	0	1	0	0	0	0	0	0	0
3	0	0	0	1	0	0	0	0	0	0
4	0	0	0	0	1	0	0	0	0	0
5	0	0	0	0	0	1	0	0	0	0
6	0	0	0	0	0	0	1	0	0	0
7	0	0	0	0	0	0	0	1	0	0
8	0	0	0	0	0	0	0	0	1	0
9	0	0	0	0	0	0	0	0	0	1

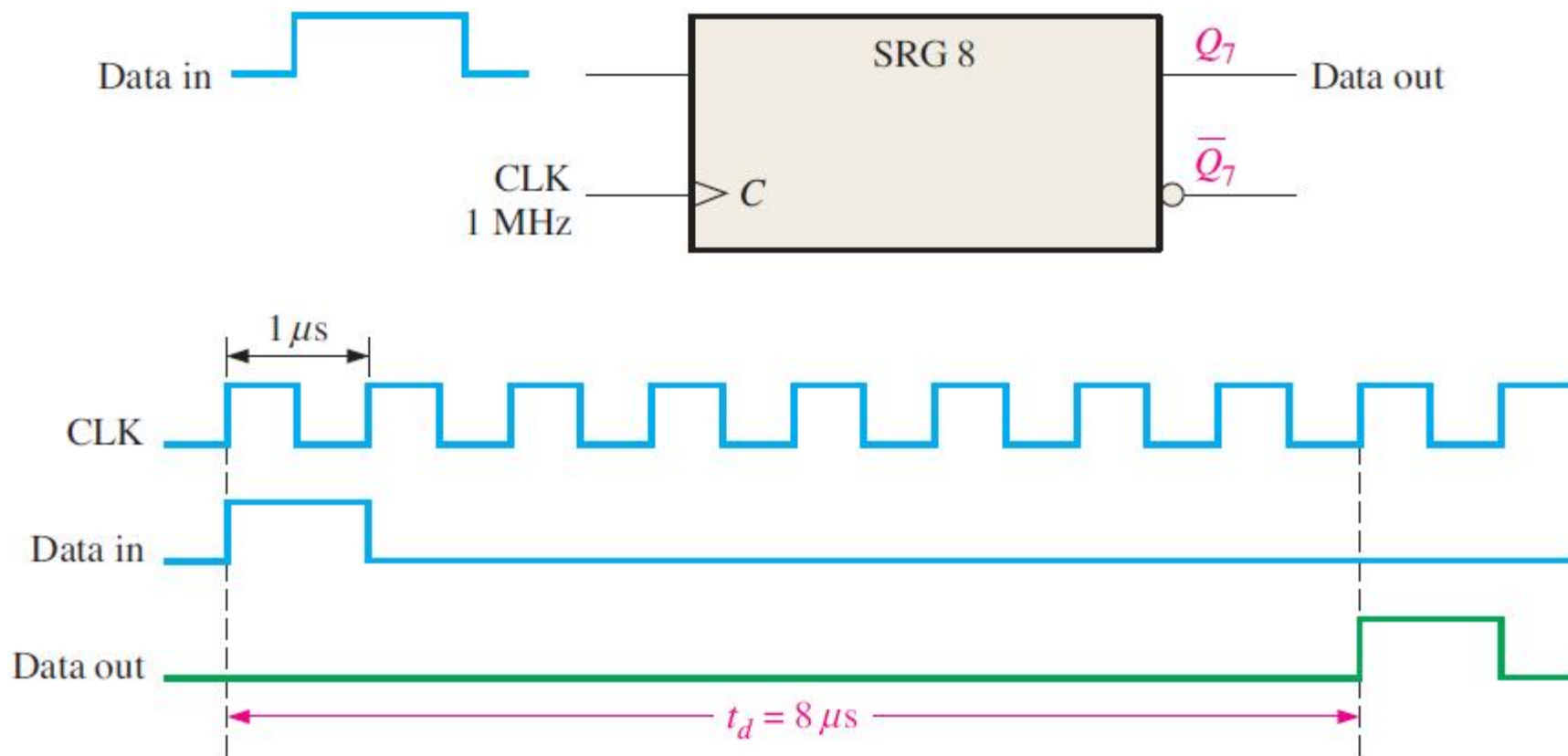
例题：一个10位环形计数器初始状态为1010000000，确定每个Q的输出波形





移位寄存器

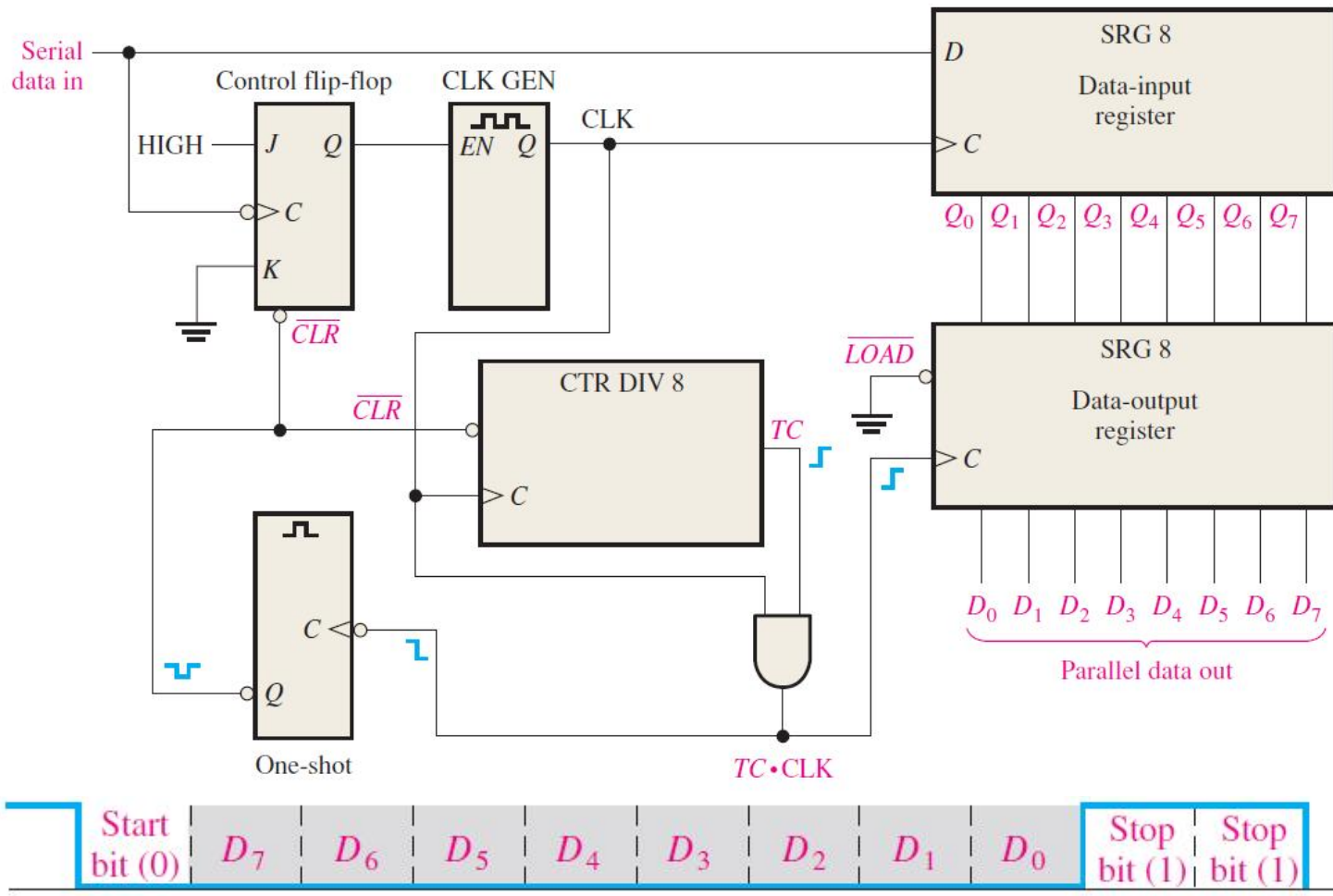
- 移位寄存器应用：时间延迟

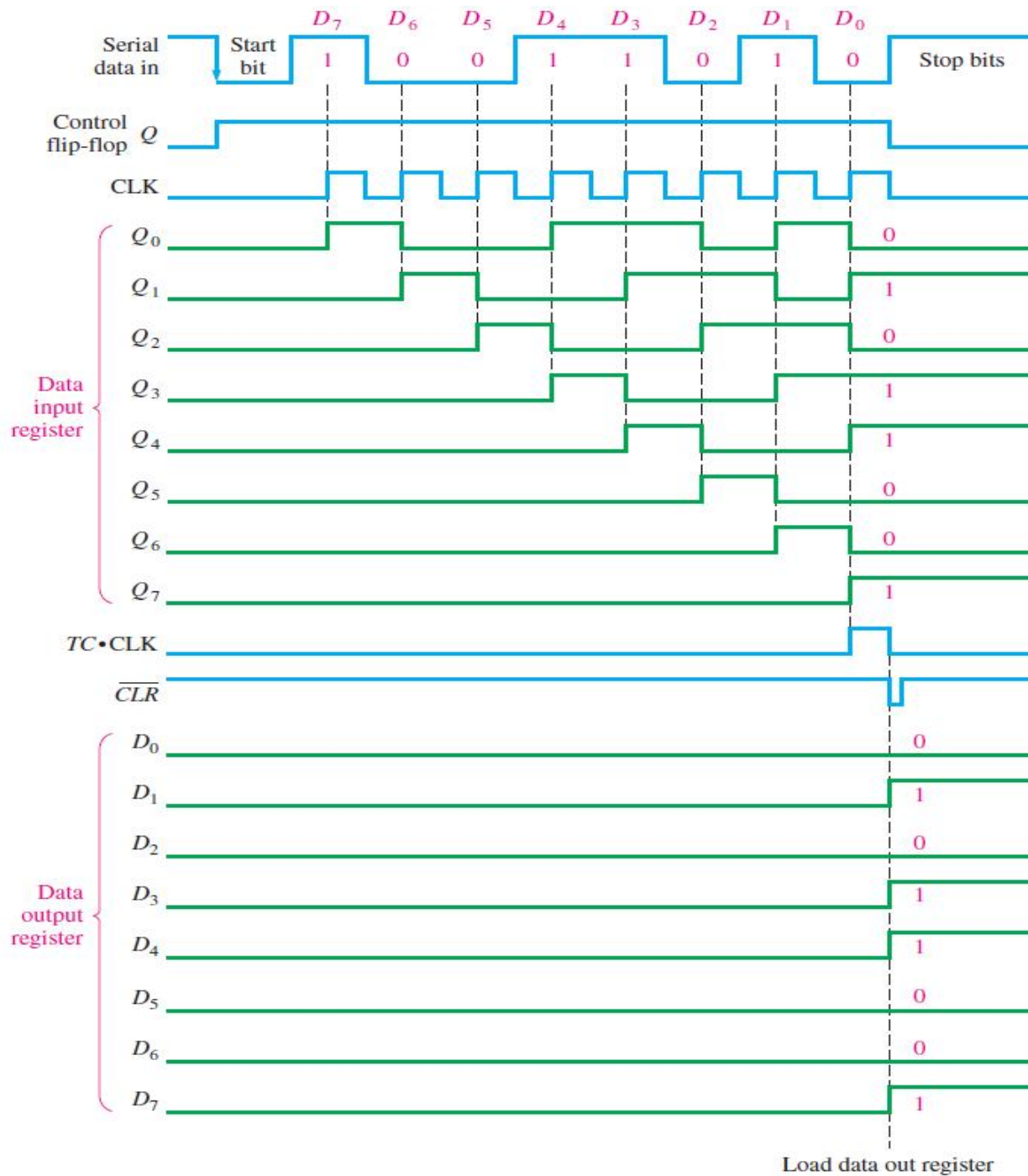




移位寄存器

- 移位寄存器应用：串行到并行数据转换

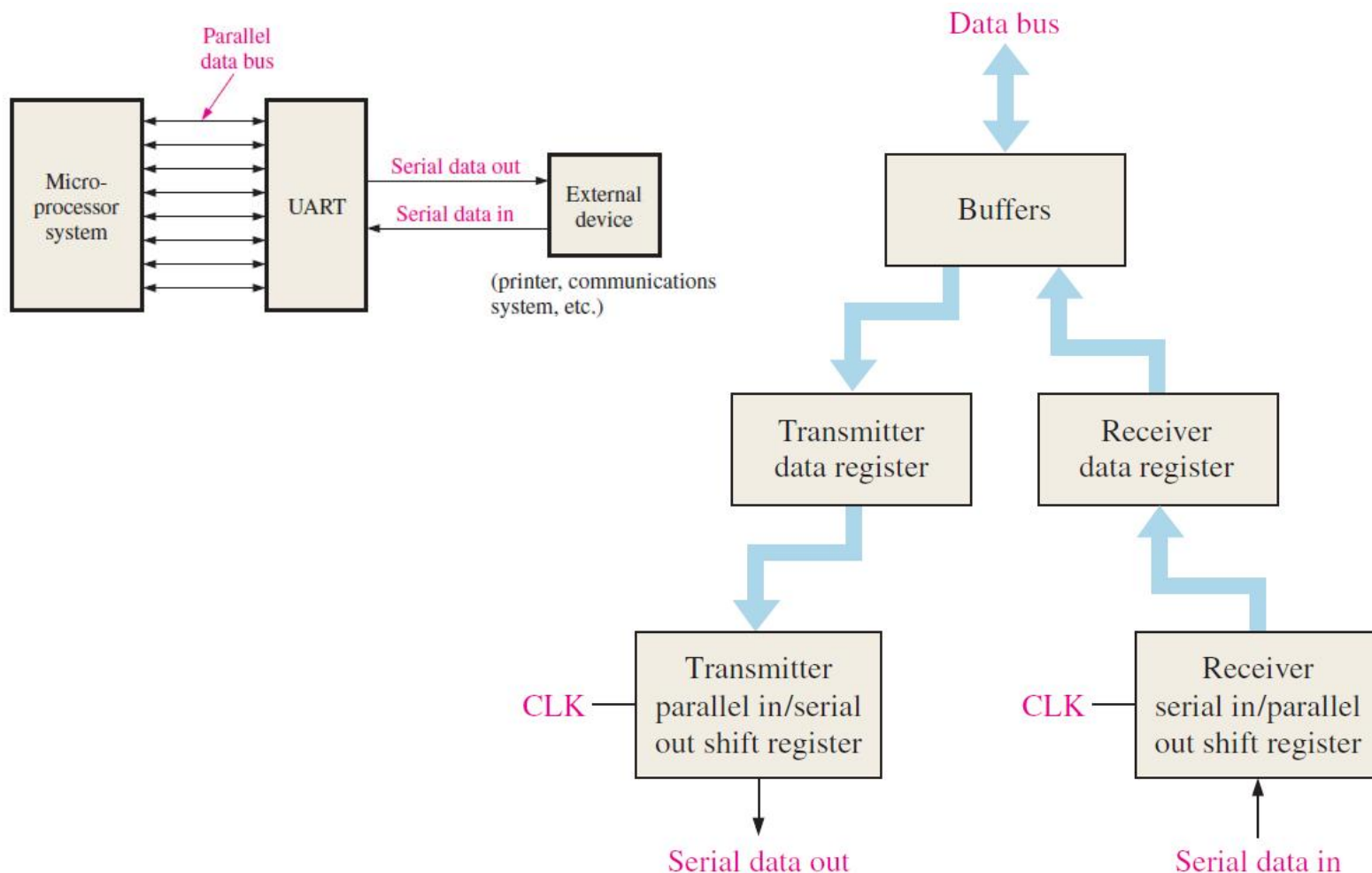






移位寄存器

- 移位寄存器应用：键盘译码器





课程回顾

- The concept of shift register
- Some kinds of shift register
 - Serial In/Serial Out
 - Serial In/parallel Out
 - Parallel In/Serial Out
 - Parallel In/Parallel Out
 - Bidirectional Shift
- Shift register counters
- Applications



作业

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